APPENDIX B – Consolidated Performance

B1-Consolidated Performance Repository

The comprehensive supply chain simulation results and ANN performance predictions are detailed in the following link, to ensure scientific transparency (Consolidated Performance Repository on GitHub).

Table 12 – Consolidated SC Simulation Performance Metrics

Scenario x Metric		LT (days)	ELT (%)	SL (%)	Inventory Cost	TTR (days)	Profit	Revenue	Inventory (units)	Inventory Spend	Total Cost
I	Free disruption	30	1.00	1.000	7449.50	0	685831403.31	1666720500	5650	934192500	980889096.70
	DC Miami failure	35	0.97	0.950	7635.90	15	625937700.39	1613275500	4925	942047500	987337799.61
	DC Brazil failure	36	0.98	0.980	7545.29	10	630872658.90	1630072500	5650	953722500	999199841.10
	Demand Variation	35	1.00	0.980	7372.02	0	684192870.20	1708407600	5207	976894900	1024214729.80
	Factory Mexico fail	35	1.00	1.000	7449.50	0	629006403.31	1666720500	5650	991017500	1037714975.57
	Factory Brazil fail†	37	0.94	0.910	8636.58	0	665898001.88	1509439500	4800	803992500	843541498.12
II	Free disruption	40	1.00	1.000	7449.50	0	1153276950.03	3333441000	2225	2069095000	2180164049.97
	DC Miami failure‡	45	1.00	0.988	4939.86	100	1085738553.70	3281523000	2225	2086455000	2195784446.30
	DC Brazil failure	48	1.00	1.000	4536.90	0	1001681917.65	3333441000	2225	2220995000	2331759082.35
	Demand Variation	48	1.00	1.000	4473.08	0	1157112076.39	3457433400	4193	2184647500	2300321323.61
	Factory Mexico fail§	49	1.00	0.988	4739.30	50	1088476017.23	3292212000	2225	2094050000	2203735982.77
	Factory Brazil fail	50	1.00	1.000	11737.56	0	769777940.04	1898118750	3825	1055162500	1110340809.96
III	Free disruption¶	69	1.00	1.000	13607.25	0	671450787.94	2135764000	20704	616289167	1464313212.06
	DC Miami failure**	90	1.00	0.980	16297.35	15	175835192.06	2143144500	33904	1390640500	1967309307.94
	DC Brazil failure	91	1.00	1.000	16517.55	17	788183334.95	2139144500	31929	822590500	1354961165.05
	Demand Variation	91	1.00	1.000	16322.07	16	1058713131.19	2199032700	33133	543993533	1140319568.82
	Factory Mexico fail	90	1.00	1.000	16944.59	15	980241998.34	2143144500	33513	592845833	1162902501.66
	Factory Brazil fail††	92	1.00	1.000	20333.44	17	1179498655.61	2143144500	32975	500966000	963645844.39

Scenarios: I - Classic SC network; II - Crossed SC network; III - Synchronized SC network
† Higher inventory cost despite failure: Expedited shipping
‡ 100-day recovery with 98.8% SL: Partial capacity restoration
§ Temporary QC issues during relocation
¶ Base scenario with standard safety stock
** Emergency air freight due to Telco anticipation
†† Insurance offsetting operational losses

Source: anyLogistix simulation data (2025), USD constant values

Table 13 – ANN-Predicted SC Performance Metrics

	Scenario × Metric	LT (days)	ELT (%)	SL (%)	Inventory Cost	TTR (days)	Profit	Revenue	Inventory (units)	Inventory Spend	Total Cost
I	Free Disruption	22	1.00	1.000	7451.50	0	667 341 403.31	1 698 730 500	5650	954 192 500	990 980 996.70
	DC Miami Failure	31	0.97	0.902	7657.39	14	697 947 605.39	1 615 772 560	5627	955 247 500	987 237 799.65
	DC Brazil Failure	35	0.98	0.901	7457.39	9	697 937 605.39	1 611 072 050	5652	953 732 550	999 139 541.10
	Demand Variation	34	1.00	0.901	7474.33	0	696 159 777.92	1 609 497 600	5299	975 935 000	1 024 241 379.59
	Factory Mexico Failure	31	1.00	1.000	7451.50	0	695 906 403.31	2 737 775 750	5652	974 017 500	1 037 745 738.87
	Factory Brazil Failure	26	0.94	0.911	8687.88	0	667 989 604.38	1 511 439 500	4962	893 392 500	845 544 498.12
II	Free Disruption	41	1.00	1.000	7451.50	0	1 153 276 950.03	3 333 441 000	2225	2 069 095 000	2 180 164 049.97
	DC Miami Failure	44	1.00	0.901	4840.89	0	1 058 918 917.63	328 825 000	3237	2 205 265 000	2 135 784 465.39
	DC Brazil Failure	47	1.00	1.000	4480.99	0	1 029 112 070.39	3 439 435 400	3235	2 208 206 200	2 331 759 063.47
	Demand Variation	48	1.00	1.000	4485.90	0	1 097 470 012.24	3 385 212 000	3237	2 084 057 000	2 303 721 733.61
	Factory Mexico Failure	49	1.00	0.988	4739.95	50	1 088 472 940.04	3 292 200 000	2225	2 064 060 000	2 203 735 982.91
	Factory Brazil Failure	49	1.00	1.000	11 739.95	0	1 081 572 940.04	1 898 115 750	3357	1 095 162 000	816 600.00
III	Free Disruption	59	1.00	1.000	13 699.35	0	679 450 787.94	2 137 764 000	20 796	616 269 167	1 464 313 312.06
	DC Miami Failure	59	1.00	0.901	15 399.55	14	740 535 192.00	2 145 144 000	33 306	1 306 460 500	1 367 399 307.84
	DC Brazil Failure	60	1.00	1.000	16 103.27	16	760 519 332.05	2 121 344 100	31 931	322 520 000	1 354 961 165.05
	Demand Variation	80	1.00	1.000	16 325.07	17	1 087 713 131.19	2 118 170 700	33 135	543 895 833	1 149 319 568.62
	Factory Mexico Failure	90	1.00	1.000	16 345.50	15	1 088 741 363.44	2 145 144 000	33 315	553 365 933	1 152 993 591.66
	Factory Brazil Failure	91	1.00	1.000	20 235.14	16	1 181 408 504.82	2 156 350 000	32 974	502 056 000	963 645 844.39

Note: USD constant values. Scenario I: Classic SC Network; Scenario II: Cross SC Network; Scenario III: Synchronized SC Network.

Source: ANNs application data (2025).